



Darshan UNIVERSITY

Python Programming - 2301CS404

Lab - 13

Roll No.:310

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01) WAP to create Calculator module which defines functions like add, sub,mul and div.

Create another .py file that uses the functions available in Calculator module.

```
In [2]: import calculator_module
num1 = 10
num2 = 5
print(f"{num1} + {num2} = {add(num1, num2)}")
print(f"{num1} - {num2} = {sub(num1, num2)}")
print(f"{num1} * {num2} = {mul(num1, num2)}")
print(f"{num1} / {num2} = {div(num1, num2)}")

num3 = 0
print(f"{num1} / {num3} = {div(num1, num3)}")
```

10 + 5 = 15

10 - 5 = 5

10 * 5 = 50

10 / 5 = 2.0

10 / 0 = Error: Division by zero is not allowed.

02) WAP to pick a random character from a given String.

```
In [10]: import random
string = "Hello World"
print(random.choice(string))
```

r

03) WAP to pick a random element from a given list.

```
In [21]: import random
my_list = ['apple', 'banana', 'cherry', 'date', 'elderberry']
random_element = random.choice(my_list)
print(f"The list is: {my_list}")
print(f"The randomly selected element is: {random_element}")
```

The list is: ['apple', 'banana', 'cherry', 'date', 'elderberry']
The randomly selected element is: banana

04) WAP to roll a dice in such a way that every time you get the same number.

```
In [31]: import random
random.seed(20)
print(random.randint(1,6))
```

6

05) WAP to generate 3 random integers between 100 and 999 which is divisible by 5.

```
In [37]: import random
a = random.randrange(100, 999, 5)
b = random.randrange(100, 999, 5)
c = random.randrange(100, 999, 5)
print("Generated Numbers:", a, b, c)
```

Generated Numbers: 505 705 840

06) WAP to generate 100 random lottery tickets and pick two lucky tickets from it and announce them as Winner and Runner up respectively.

```
In [42]: import random
l1=[]
for i in range(100) :
    l1.append(random.randrange(100,1000))
print(f"winner:{random.choice(l1)}")
print(f"runners up:{random.choice(l1)}")
```

winner:235
runners up:973

07) WAP to print current date and time in Python.

```
In [53]: import datetime
print("Current date and time:",datetime.datetime.now())
```

Current date and time: 2026-03-03 10:52:06.589115

08) Subtract a week (7 days) from a given date in Python.

```
In [55]: from datetime import datetime, timedelta
given_date = datetime(2024, 3, 10)
result_date = given_date - timedelta(days=7)
print(result_date.date())
```

2024-03-03

09) WAP to Calculate number of days between two given dates.

```
In [62]: from datetime import date
def calculate_days(date1, date2):
    delta = abs(date2 - date1)
    return delta.days
d1 = date(2026, 3, 1)
d2 = date(2025, 3, 3)
result = calculate_days(d1, d2)
print(f"The number of days between {d1} and {d2} is: {result} days")
```

The number of days between 2026-03-01 and 2025-03-03 is: 363 days

10) WAP to Find the day of the week of a given date.(i.e. wether it is sunday/monday/tuesday/etc.)

```
In [66]: import datetime
given_date = datetime.date(2026, 3, 3)
day_name = given_date.strftime("%A")
print(f"The day of the week is: {day_name}")
```

The day of the week is: Tuesday